

Custodly

The collateral truth layer for robot lending. This document is the build plan: architecture, integration approach, phased milestones, and go-to-market sequencing, meant to be worked from directly, not pitched from.

1. SYSTEM ARCHITECTURE

Four layers. Each one is intentionally small in scope for a two-person team; the goal is a working, narrow version of each before expanding brand coverage or feature depth.

- **1. Robot connectors:** brand-specific integration modules that pull telemetry (uptime, location, battery/utilization, fault codes) from each manufacturer's API or cloud platform. Start with one brand only.
- **2. Ingestion + normalization layer:** a common schema that maps every brand's differently-shaped data into one consistent 'robot health record,' so the dashboard and risk logic don't need to know which brand they're looking at.
- **3. Risk scoring engine:** rules-based at first (e.g. offline > 72 hours, utilization drop > 40% week-over-week, repeated fault codes), not machine learning. Machine learning only makes sense once there's enough historical default/repossession data to train against, which won't exist for a long time.
- **4. Lender dashboard + alerting:** a simple web view per lender showing every financed robot they have collateral exposure to, its current status, and a flag list. Email/SMS alerts on flag triggers, nothing more elaborate needed for v1.

2. DATA ACCESS MODEL

The core defensibility of this business is that data access is a loan condition, not a scrape or a public API integration a competitor could copy overnight.

- **Consent flow:** borrower authorizes robot-account API access to Custodly as part of loan closing, the same pattern as connecting a bank account via Plaid during a loan application, or a telematics device consent in equipment/auto lending.
- **Legal language:** needs real loan-covenant wording, not improvised. This is explicitly flagged as a next step requiring someone who has actually underwritten equipment finance deals, not something to draft from assumption.
- **Data ownership:** lender is the customer and data recipient; borrower is the data source; Custodly is the pipe and scoring layer in between. No party other than the lender should see another lender's portfolio data.

3. PHASED BUILD PLAN

Four phases, roughly six months to a real design-partner pilot. Timeframes assume part-time work around school and existing jobs, so treat weeks as sequencing, not hard deadlines.

Phase 0: Foundation // Weeks 1-3

Goal: *Confirm the single hardest technical unknown, whether a real robot brand's API actually exposes usable telemetry, before building anything else.*

- Samuel evaluates Pudu's developer API/docs directly; confirm what data is actually exposed (not just what marketing claims).
- If Pudu's API is unusable or access-gated, fall back to Bear Robotics or Keenon; the point of this phase is picking one that is actually buildable against.
- Antonio drafts the plain-language consent/authorization flow concept (not legal language yet) to react to once real API scope is known.

Phase 1: Single-Brand Prototype // Weeks 4-9

Goal: *One robot, one brand, one lender-facing dashboard screen. Nothing else. This is the artifact that makes every future conversation credible.*

- Build the connector for the chosen brand and the normalization schema (even if it only has to normalize one brand for now, structure it so a second brand can be added without a rewrite).
- Build the rules-based flag logic: offline duration, utilization drop, fault code recurrence.
- Build a bare dashboard: robot list, status, flag history. No polish needed yet, it needs to work end-to-end on real or realistic test data.

Phase 2: Design Partner // Weeks 10-18

Goal: *Get one real lender or financing broker using this on at least one real financed robot, even informally and unpaid.*

- Target small equipment-finance lenders or brokers already financing service robots, not large banks; a small shop can move fast and give direct feedback.
- Formalize the consent/covenant language with actual input from someone in equipment finance underwriting.
- Run the pilot on one real robot if at all possible; a live example is worth more than ten more mockups.

Phase 3: Expand + Raise // Weeks 19-26

Goal: *Add a second robot brand, tighten the pitch with real pilot data, and start outreach to early-stage funds.*

- Add the second brand connector once the first is proven; this is also the point where the normalization schema gets validated as actually reusable.
- Replace every projected number in the pitch deck with real pilot data where available.
- Begin outreach using the existing VC/PE tracker, filtered to early-stage, robotics- or fintech-adjacent funds that write pre-seed/seed checks.

4. GO-TO-MARKET SEQUENCING

The buyer is the lender, not the robot owner. This changes the sales motion from typical SaaS: fewer, larger, longer-cycle relationships, not high-volume self-serve signups.

- **Who to sell to first:** small and mid-size equipment finance lenders and brokers already active in robotics/automation financing, not large banks who move slowly and want vendor track record Custodly won't have yet.
- **Wedge:** offer the pilot free or near-free in exchange for real usage data and a case study; the goal in year one is proof, not revenue.
- **Expansion path:** once one lender's pilot works, the natural next step is other lenders financing the same robot brands, since the connector work is already done.
- **Long-run model:** per-robot-per-month monitoring fee paid by the lender, referenced against comps like CDK Global and Solera in the automotive/equipment space (see pitch deck, Market slide).

5. OPEN QUESTIONS TO RESOLVE EARLY

- Samuel's exact technical scope and robotics-specific experience, still not fully defined; this is the single largest risk to execution feasibility.
- Whether robot manufacturers will even grant API access to a third party for this purpose, versus treating it as a data-partnership conversation the manufacturer itself needs to approve.
- Realistic scale of small-business robot financing volume today; this needs a real number before sizing outreach or fundraising targets, not an assumption.

This blueprint reflects working judgment from a single planning session, not a finalized technical spec or legal document. Treat every timeline, comp, and access assumption as directional, and re-verify against real API documentation, lender conversations, and legal counsel before committing resources.